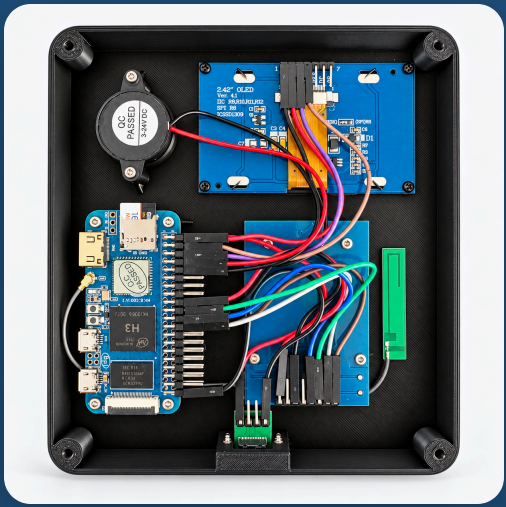
ICUBE DEVELOPMENT (CALGARY) LTD.

Timeclock Build and Install Guide

From case files to a working clock on the wall: print, assemble, prepare the image, upload the software and install it.





Case v8.34 · Image bpi-zero-wbuild 3.13 · mk-timeclock Lite 1.9.5-RC3-cleanup6

For IT and installers

START HERE

Overview and what you need

1. **Case files:** use the supplied STL files, or regenerate them from the OpenSCAD source.
2. **Print** the front, the rear cover and an OLED fit test.
3. **Assemble** the electronics into the front and wire them.
4. **Prepare the image:** flash the microSD card and set Wi-Fi and the root password.
5. **Upload** the timeclock package to the clock over the network.
6. **Install** the timeclock software, then close the case and mount it.

Downloads

- Case STLs and source: `bpi_rfid_timeclock_v8_34_two_piece.scad`
- Image release 3.13: *github.com/mkinderwater/bpi-zero-wbuild/releases/tag/3.13*
  - `bpi-zero-wbuild-3.13-trixie-minimal-hw-bpi-m2-zero.img.gz`
  - `…img.gz.sha256` (checksum)
- Timeclock package: `mk-timeclock-lite-RC3-cleanup6.zip`

Tools

- FDM 3D printer, 0.4 mm nozzle
- Small Phillips screwdriver
- Soldering iron and solder, for the RC522 and USB-C header pins and the buzzer leads
- A computer with a microSD card reader
- An SD flashing tool that reads `.img.gz` files (balenaEtcher, Raspberry Pi Imager or similar)
- An SSH client: Terminal (macOS/Linux) or Windows Terminal/PowerShell, plus WinSCP on Windows

Network

- A 2.4 GHz Wi-Fi network the clock can join (name and password)
- Internet access during installation, for software packages
- Access to the router or DHCP server, to find the clock's address and reserve it

> **Keep the clock on a trusted network**
> The timeclock is designed for a private company network. Don't expose it to the internet.

Roles

One clock is the **Primary**. It holds the database and the manager website. Any other clocks are **Secondary**. They record taps and send them to the Primary. Build and install the Primary first.

START HERE, CONTINUED

Parts list

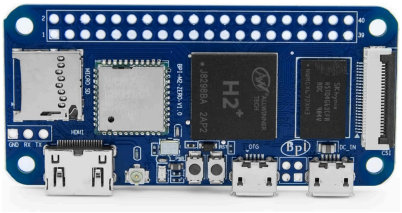
PART	QTY	NOTES
Banana Pi BPI-M2 Zero	1	65 × 30 mm board, 40-pin header fitted
microSD card	1	16 GB or larger, good-quality brand
2.42" SSD1309 OLED, 128 × 64	1	PCB 71.13 × 43.57 mm, glass 62.2 × 39 mm, set for I2C (IIC) at address 0x3C
RC522 RFID reader (13.56 MHz)	1	PCB about 59.4 × 39.9 mm, header fitted
Active buzzer, 22.6 mm round	1	Must work directly from a 3.3 V GPIO pin, or use a transistor driver
USB-C power breakout	1	20 × 7 mm PCB, 16 mm hole spacing. Easiest: power it with a USB-A to USB-C cable, which needs nothing extra. For a USB-C to USB-C charger, the breakout needs 5.1 kΩ pull-down resistors on CC1 and CC2. No PD trigger board
Wi-Fi antenna, IPEX/u.FL	1	Flat stick-on antenna for the board's antenna connector
Female-female jumper wires	13	Dupont type, 10 to 20 cm
M2 × 6 mm self-tapping screws	4	Banana Pi
M2 × 5 mm self-tapping screws	8	OLED (4), RC522 (4)
M2 × 6 mm self-tapping screws	2	USB-C breakout, fitted from outside
M2 × 8 mm self-tapping screws	2	Buzzer (through its ears or the printed clamp)
M3 × 12 mm countersunk self-tapping screws	4	Rear cover
5 V USB power supply	1	2 A or more. A USB-A charger with a USB-A to USB-C cable is simplest. Plain 5 V only
Wall screws	2	Pan head, head up to 8 mm and shank up to 4.3 mm, with anchors to suit the wall

Screw lengths are worked out from the case model (pilot depths and post heights). Check them against your hardware before a production run.

START HERE, CONTINUED

Components

Photos of the reference parts. Prepare each one as described before assembly.



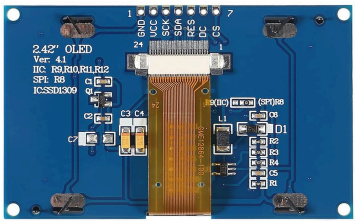
Banana Pi BPI-M2 Zero

Pin 1 is the square pad at the end nearest the microSD slot. Odd pins (1 to 39) are on the inner row and even pins (2 to 40) are along the board edge. The Wi-Fi antenna clips onto the small round connector beside the HDMI port.



RC522 reader

Solder the supplied 8-pin header to the row labelled SDA, SCK, MOSI, MISO, IRQ, GND, RST, 3.3V. The card and key fob in the kit are useful for testing.



2.42" SSD1309 OLED, back

Connector pins 1 to 7: GND, VCC, SCK (I2C clock), SDA, RES, DC, CS. Convert it to IIC mode as shown below.



OLED, front

The raised glass with its stepped lower edge fits the case window. Handle it by the PCB edges.



Active buzzer

Red lead is + (to pin 7), black is − (to pin 6). Crimp female Dupont housings onto the bare leads, or solder them to jumper wires. The two ears take the M2 × 8 mm screws.



USB-C breakout

Pads G, D+, D−, V. Use only **V** (5 V, to pin 2) and **G** (ground, to pin 39). D+ and D− stay unused. Solder header pins or wires to V and G. The white plate with two holes screws into the case pocket.

Set the OLED to I2C (IIC) mode

Most modules ship set for SPI. Change these positions on the back of the OLED (printed beside the connector and in the "IIC: R9, R10, R11, R12 / SPI: R8" note) before assembly:

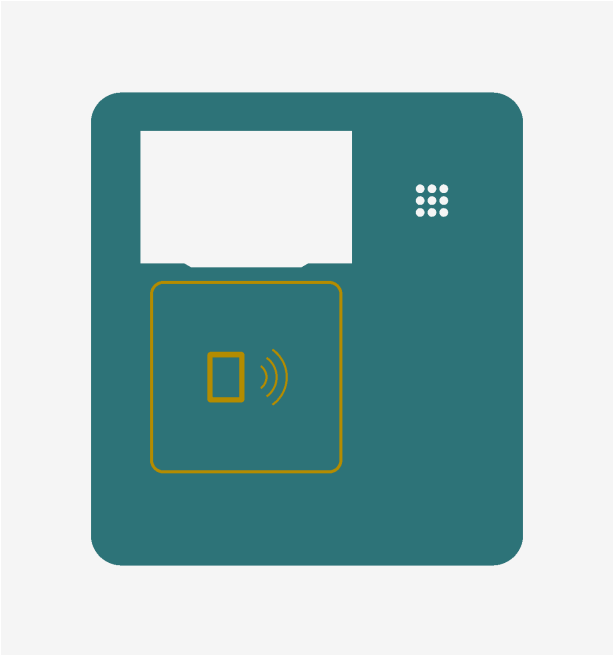
POSITION	CHANGE
R8	Remove the 0 Ω resistor, leaving the pads open
R9	Fit the removed resistor here, or solder-bridge the pads
R10	Solder-bridge
R11	Solder-bridge
R12	Solder-bridge

CHAPTER 1

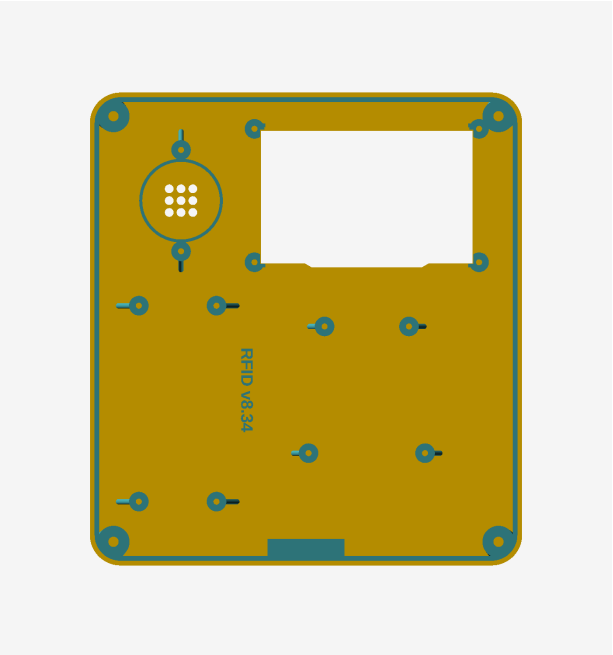
Case files: SCAD to STL

The case is one OpenSCAD file. A setting named `part` chooses which piece to make. The STL files supplied with this guide were made from v8.34 with OpenSCAD and checked as closed solids.

FILE	PART =	WHAT IT IS
`timeclock_v8_34_front_PRINT.stl`	"front"	Front shell with display window, tap zone, buzzer vents and all mounting posts. Already flipped face-down for printing.
`timeclock_v8_34_back.stl`	"back"	Rear cover with vents, two wall keyholes and four countersunk screw holes.
`timeclock_v8_34_oled_fit_test.stl`	"oled_fit_test"	Small coupon cut from the real front, to check the OLED fit before the full print.
`timeclock_v8_34_buzzer_clamp.stl`	"buzzer_clamp"	Optional bar to hold a buzzer that has no mounting ears.



Front, outside



Front, inside (seen from the open back)

To regenerate an STL

1. Open the `.scad` file in OpenSCAD (2021.01 or newer).
2. Near the top, set `part = "front";` (or "back", "oled_fit_test", "buzzer_clamp").
3. Press F6 to render, then use **File > Export > Export as STL**.

From a command line: `openscad -o front.stl -D 'part="front"' bpi_rfid_timeclock_v8_34_two_piece.scad`. The exported front is face-up, so turn it face-down in the slicer. Don't change dimensions unless you have measured your own parts. The posts and window match the reference parts listed earlier.

CHAPTER 2

Printing

1. Print the OLED fit test first

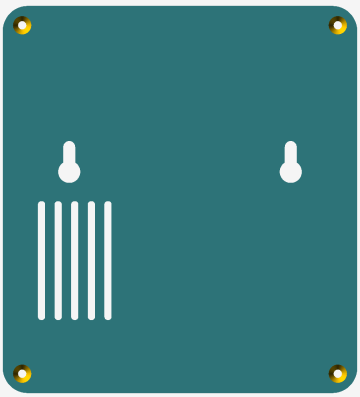
It takes a fraction of the material. Press the OLED glass into the window from the back. It should sit flush with the front face, with the PCB resting on the four corner seats and the screw holes lining up. If it's tight, check your printer's dimensional accuracy before printing the front.



OLED fit test coupon

2. Print the front and rear cover

PART	ORIENTATION	SUPPORTS
Front	Visible face down on the bed (the _PRINT file is already this way)	None. Corner screw bosses are designed to grow from the walls without supports.
Rear cover	Outer face down, rim up	None
Fit test, clamp	Flat, as exported	None



Rear cover, outside: vents, keyholes, countersunk holes

Suggested settings

- Layer height 0.24 mm (the model's support-free features assume this), 0.4 mm nozzle
- PETG or PLA in black. A textured bed sheet gives the finish shown on the cover.
- 3 or more walls, 20 to 30 % infill

After printing

- Remove any stringing inside the window and buzzer vents.
- Check the rear cover fits over the front's stepped rim all the way round. The overlap blocks light leaking through the seam.

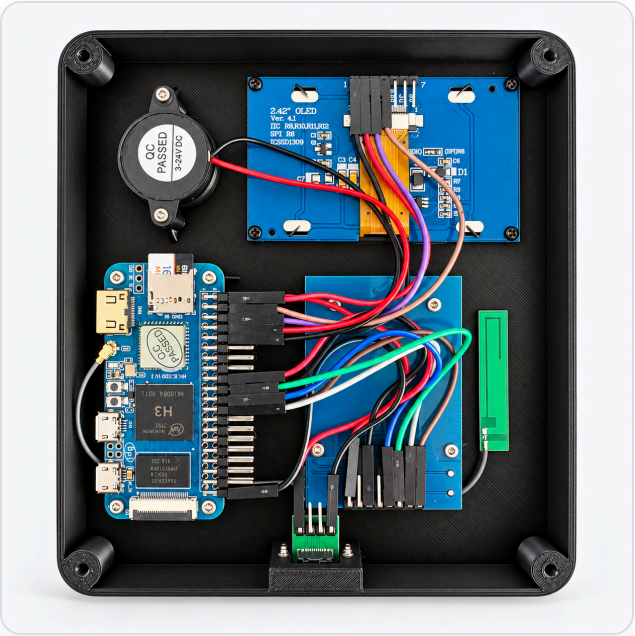
CHAPTER 3

Assembly

Work with the front lying face-down and the open back facing you. Everything mounts on posts moulded into the inside of the front. All screws are self-tapping, so turn them in gently and stop when snug.

1. **USB-C breakout.** From outside, seat the board in the pocket on the bottom edge so its face is flush with the case. Fix it with 2 × M2 × 6 mm. Its pins point into the case.
2. **OLED.** Make sure the module is in I2C (IIC) mode. Convert it first: R8 open, R9 fitted or bridged, R10 to R12 bridged (see Components). Press the glass into the window from inside, so the PCB sits on the four corner seats. Fix it with 4 × M2 × 5 mm. The screws go into blind holes and don't come through the front.
3. **RC522 reader.** Antenna side toward the front, on the four short posts behind the tap-zone outline. Fix it with 4 × M2 × 5 mm.
4. **Buzzer.** Seat it in the round locating ring behind the vent holes. Fix it with 2 × M2 × 8 mm through its ears, or with the printed clamp if it has no ears.
5. **Banana Pi.** On the four posts in the lower corner, header toward the middle of the case. Fix it with 4 × M2 × 6 mm.
6. **Wi-Fi antenna.** Clip it onto the board's antenna connector and stick it to the inside wall, away from the RC522.
7. **Wiring.** Connect everything as shown on the next page.

> **Leave the rear cover off for now**
> The microSD slot is reached from inside. Fit the cover after the software checks pass (Chapter 6).

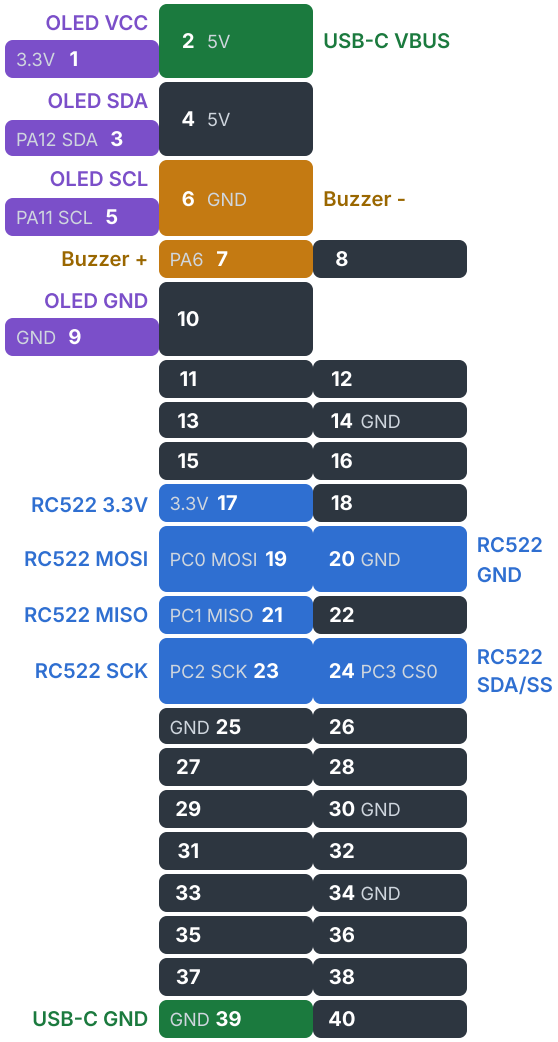


Reference build, from the open back

Wiring

Pin numbers are the **physical pin numbers** on the Banana Pi's 40-pin header. Pin 1 is the square pad, at the end nearest the microSD slot. Odd pins are on one row and even pins on the other. Unplug power while wiring.

FROM	TO HEADER PIN
RC522 reader (SPI0)	
SDA / SS	24 (PC3, SPI0 CS0). Despite the SDA label, this is the SPI chip select
SCK	23 (PC2, SPI0 clock)
MOSI	19 (PC0), data from the board to the reader
MISO	21 (PC1), data from the reader to the board
IRQ	Not connected (the software polls the reader)
GND	20 (GND)
RST	Not connected (not used)
3.3V	17 (3.3 V). **Never power the RC522 from 5 V**
OLED (I2C0, address 0x3C)	
1 GND	9 (GND)
2 VCC	1 (3.3 V)
3 SCK (SCL)	5 (PA11)
4 SDA	3 (PA12)
5 RES, 6 DC, 7 CS	Leave unconnected **only if** the module's onboard circuitry sets them in I2C mode, as on the reference module. Check your module's documentation
Buzzer (active)	
+ (red)	7 (PA6, GPIO offset 6). For a 3-terminal buzzer module, pin 7 goes to its signal terminal and power follows its rating
– (black)	6 (GND)
USB-C power breakout	
V (VBUS, 5 V)	2 (5 V rail), regulated 5 V input
G (GND)	39 (power ground)



Leave all other header pins unconnected. Some are reserved by the image for audio. The software uses /dev/spidev0.0 for the reader and /dev/i2c-0 at address 0x3C for the display.

Check twice before powering on

5 V on a 3.3 V pin or a reversed buzzer can damage the board. A buzzer wired straight to pin 7 must be GPIO-compatible (3.3 V, low current). Otherwise drive it through a transistor.

CHAPTER 4

Prepare the image

The clock runs a purpose-built Debian 13 image for the BPI-M2 Zero, release 3.13. It has SPI, I2C and GPIO ready for the timeclock and boots headless (no monitor needed).

1. Download and check

From the 3.13 release page, download `bpi-zero-wbuild-3.13-trixie-minimal-hw-bpi-m2-zero.img.gz` and its `.sha256` file into the same folder. Check that the download is complete:

- **macOS / Linux:** `shasum -a 256 -c bpi-zero-wbuild-3.13-trixie-minimal-hw-bpi-m2-zero.img.gz.sha256` should print **OK**.
- **Windows (PowerShell):** `Get-FileHash .\bpi-zero-wbuild-3.13-trixie-minimal-hw-bpi-m2-zero.img.gz`, then compare the hash with the text in the `.sha256` file.

2. Flash the microSD card

Write the `.img.gz` file directly with a tool that supports compressed images, such as balenaEtcher or Raspberry Pi Imager ("Use custom"). You don't need to unzip it first. Flashing erases the card.

3. Set Wi-Fi and the root password

After flashing, remove and reinsert the card. A small drive named **BPIWBUILD** appears. Open `CONFIG.TXT` on it in a plain text editor and fill in:

```
SSID=YourWiFiName
PSK=YourWiFiPassword
COUNTRY=CA
HIDDEN=false
TIMEZONE=America/Edmonton
ROOT_PASSWORD=a-strong-password-8-to-64-chars
```

- **ROOT_PASSWORD is required** (8 to 64 characters). There is no default login, and without it the clock can't be reached.
- PSK is 8 to 63 characters. Set HIDDEN=true only for a hidden network.
- Use your local time zone, for example America/Vancouver or America/Toronto.

Save, eject the card safely, and put it in the Banana Pi's microSD slot.

> **Your passwords are removed after first boot**
> Once setup succeeds, the clock blanks PSK= and ROOT_PASSWORD= in CONFIG.TXT. Record the root password in your password manager.

First boot

1. Plug the power supply into the case's USB-C port. A USB-A charger with a USB-A to USB-C cable always works. The board's red LED lights.
2. Wait several minutes. On first boot the clock expands the card, sets the root password, creates its SSH keys, installs Wi-Fi support and joins your network.
3. Find the clock's IP address in your router's or DHCP server's client list. Each clock has its own hostname. If a monitor is attached to the board's mini-HDMI port, the login screen shows **Wi-Fi: wlan0 MAC: … IPv4: …**.
4. Reserve that IP address for the clock (DHCP reservation), so it never changes.
5. From your computer, sign in: `ssh root@192.168.1.50` (use your clock's address and the ROOT_PASSWORD you set).
6. Confirm the image: `cat /etc/bpi-zero-wbuild-release` should show version **3.13-trixie-minimal-hw**.

> **If you can't connect**
> The usual causes are a wrong Wi-Fi name or password, a 5 GHz-only network, or a ROOT_PASSWORD that's too short. Power off, fix CONFIG.TXT on another computer and boot again. With a monitor attached, the boot messages show which step failed.

Changing Wi-Fi later

Wi-Fi profiles are stored under `/var/lib/iwd/` on the clock. Ask your iCube contact before moving a clock to a different network.

CHAPTER 5

Upload the timeclock package

Copy `mk-timeclock-lite-RC3-cleanup6.zip` to the clock's `/root` folder.

macOS, Linux or Windows PowerShell (scp)

From the folder that contains the zip file:

```
scp mk-timeclock-lite-RC3-cleanup6.zip
root@192.168.1.50:/root/
```

Enter the root password when asked. The first time, type **yes** to accept the clock's key.

Windows (WinSCP)

1. New site: **SFTP**, host name = the clock's IP, port 22, user **root**, and the root password.
2. Select **Login** and accept the host key.
3. On the right (clock) side, open `/root`. Drag the zip file across.

Unpack it on the clock

Sign in with SSH (`ssh root@192.168.1.50`) and run:

```
apt-get update
apt-get install -y unzip python3
cd /root
unzip mk-timeclock-lite-RC3-cleanup6.zip
cd mk-timeclock-lite-RC3-cleanup6
python3 rc1-preflight.py
```

The preflight check should print **PASS: source and referenced installer files present**. Use `unzip` rather than other extract tools, because it keeps the installer scripts executable.

You're signed in as root, so the commands don't need `sudo` . The image doesn't include it.

CHAPTER 6

Install the timeclock

Primary clock (install this one first)

```
cd /root/mk-timeclock-lite-RC3-cleanup6
./install-master.sh
```

The installer checks the platform, installs packages, builds the software on the clock, sets up HTTPS with a temporary certificate and starts the services. It takes a while on the first run. It finishes with **Installed role: primary**.

Secondary clocks

```
cd /root/mk-timeclock-lite-RC3-cleanup6
./install-slave.sh
```

It finishes with **Slave installed. Adopt at http://<clock-ip>:8090/**. Then on the Primary's website, go to **Device > Clock management** and select **Generate code**. Enter the code on the new clock's setup page within 10 minutes, then approve the request on the Primary.

> **Choose the role carefully**
> Never run both installers on the same clock. A clock's role can't be changed by running the other installer.

Create the first manager (Primary only)

1. In a browser, open `https://<clock-ip>/setup`. The temporary certificate causes a browser warning. Continue to the site.
2. Enter the manager's name, email, position and a password of 12 or more characters.
3. When asked, **tap a new, unused card on the clock within 30 seconds**. That card becomes the manager card. Keep a separate card for attendance.
4. Sign in at `https://<clock-ip>/login`. The Manager Guide covers everything from here.

Install a trusted HTTPS certificate

Links in employee emails need a real certificate. Under **Settings > HTTPS certificate**, upload the certificate, intermediate chain and private key for the clock's hostname. The clock checks them before switching over. Also set **Settings > Outgoing email** so notices can be sent.

Check, close and mount

Check the hardware

- After a restart the screen shows the boot logo for about two seconds, then the time and **TAP YOUR CARD**.
- Tap the manager card: the screen shows **MANAGER / NO PUNCH RECORDED** and the buzzer sounds.
- Tap an unregistered card: **INVALID CARD / NOT RECORDED**. It appears under **Today > Unknown cards**.
- If taps show **CLOCK NOT SYNCED**, the clock can't reach a time server yet. Check its internet access, or set a server under **Settings > Time sync**.

For a full service report, run `./verify-stage.sh` from the package folder over SSH.



Ready for taps

Close the case

Fit the rear cover over the stepped rim and fix it with 4 × M3 × 12 mm countersunk screws. They sit flush with the back.

Mount on the wall

1. Fit two pan-head screws, **80 mm apart** and level, leaving the heads about 3 mm out from the wall.
2. Place the clock so the screw heads go through the large openings of the two keyholes, then slide it **down** about 9 mm until it locks.
3. Route the USB-C cable down from the bottom port.

Mount it at a comfortable height, out of direct sunlight and away from large metal surfaces, which weaken the card reader.



Rear: keyholes, vents and USB-C cable

REFERENCE

Troubleshooting

PROBLEM	WHAT TO CHECK
Can't find the clock on the network	CONFIG.TXT values (SSID, PSK, 2.4 GHz). ROOT_PASSWORD present and 8 to 64 characters. Wait several minutes on first boot. Attach a monitor to read the boot messages.
ssh: Permission denied	Sign in as **root** with the ROOT_PASSWORD you set. It was blanked from CONFIG.TXT after first boot, but it's still the password.
Preflight or installer says "BPI image marker missing" or "Expected Debian armhf"	The card wasn't flashed with the 3.13 image. Reflash it.
Installer says "Missing /dev/spidev0.0" (or i2c-0, gpiochip0)	Wrong image or a damaged card. Reflash with the 3.13 image.
"Permission denied" running ./install-master.sh	The zip was extracted without keeping file permissions. Delete the folder and unpack again with `unzip`.
Screen blank, cards work	OLED wiring (pins 1, 3, 5, 9). Check its I2C configuration matches the reference module, or whether your module needs RES held high.
Cards never read	RC522 wiring (pins 17 to 24): 3.3V on pin 17, RST and IRQ not connected. Antenna facing the front. No metal behind the tap zone.
No sound	Buzzer polarity (+ to pin 7, – to pin 6). It must be an active, GPIO-compatible buzzer.
Board resets or won't start	Plain 5 V supply of 2 A or more. Check the USB-C breakout wiring (VBUS to pin 2, GND to pin 39) If a USB-C to USB-C charger won't power it, use a USB-A charger and USB-A to USB-C cable, or a breakout with 5.1 kΩ CC pull-downs.
Taps show CLOCK NOT SYNCED	Internet access for time sync, or a local time server under Settings > Time sync.
Browser certificate warning	Expected until a trusted certificate is installed (Chapter 6).

> **Record each clock**
> For every clock, keep its role (Primary or Secondary), name, IP reservation, Wi-Fi MAC address and install date.